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Perspective

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Machine Learning for Accelerated Discovery of Solar Photocatalysts

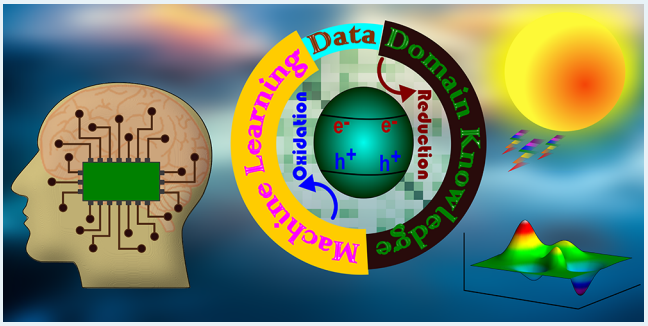
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Supporting Information

ABSTRACT: Robust screening of materials on the basis of structure−property−activity relationships to discover active photocatalysts is a highly sought out aspect of photocatalysis research. Recent advancements in machine learning offer considerable opportunities to evolve photocatalysts discovery practices. Machine learning has largely facilitated various areas of science and engineering, including heterogeneous catalysis, but adaptation of it in photocatalysis research is still at an elementary stage. The scarcity of consistent training data is a major bottleneck, and we foresee the integration of photocatalysis domain knowledge in mainstream machine learning protocols as a viable solution. Here, we present a holistic framework incorporating machine learning and domain knowledge to set directions toward accelerated discovery of solar photocatalysts. This Perspective begins with a discussion on domain knowledge available in photocatalysis which could potentially be leveraged to liaise with machine learning methods. Subsequently, we present prevalent machine learning practices in heterogeneous catalysis tailored to assist discovery of photocatalysts in a purely data-driven fashion. Lastly, we conceptualize various strategies for complementing data-driven machine learning with photocatalysis domain knowledge. The strategies involve the following: (i) integration of theoretical and prior empirical knowledge during the training of machine learning models; (ii) embedding the knowledge in feature space; and (iii) utilizing existing material databases to constrain machine learning predictions. The aforementioned human-in-loop framework (leveraging both human and machine intelligence) could possibly mitigate the lack of interpretability and reliability associated with data-driven machine learning and reinforce complex model architectures irrespective of data scarcity. The concept could also offer substantial benefits to photocatalysis informatics by promoting a paradigm shift away from the Edisonian approach.



KEYWORDS: *photocatalysis, machine learning, domain knowledge, photocatalyst discovery, high-throughput, database*

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1. INTRODUCTION

Solar-induced chemical processes, including the generation of solar fuels, are among the most critical technologies that would contribute toward achieving sustainability in the 21st century. In this regard, semiconductor photocatalysts that are capable of activation under solar or visible light (the largest component of solar spectrum) are being intensively investigated for applications ranging from water splitting,[1] carbon dioxide reduction,[2] nitrogen fixation,[3] chemicals syntheses,[4] to environmental remediation.[5,6] The urgencies in uncovering viable photocatalysts for these targeted reactions, against the backdrop of the global warming agenda, determine their potentials for industrial implementations. For example, meeting the targeted 10% solar-to-hydrogen (STH) efficiency for water splitting needs to be achieved urgently for the technology to stay relevant.[7] The same is true for the other solar-to-chemical applications, although the thresholds for potential commercialization have not been as clearly defined as the solar hydrogen generation. In all cases, however, the discovery of photocatalytic materials capable of meeting the desired conversion targets has been a persisting bottleneck.

The implementation of machine learning (ML) in the materials space has in recent years seen preliminary success across different fields, for example, in electrocatalysis,[8] perovskite solar cells, rechargeable batteries,[9] and piezoelectrics.[10] For heterogeneous catalysis, the experimental applications started to surface in the early 1990s, and the trend shaped toward computational applications in the past decade. Unlike the fields of image and speech recognition which are at the forefront of applied ML and where very large data sets are available for training ML models, the time-cost as well as complexities associated with materials synthesis and assessments severely limit the size of training data sets and

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consequently the implementation of ML in heterogeneous catalysis.

In photocatalysis, amalgamation efforts with ML are still at an elementary stage, inundated with predictive models centering the application area of water remediation.[11,12] Bracketed scientific questions, such as optimization of process parameters, determining significance of input parameters, or prediction of process outcome under known inputs are typically "low-hanging fruit". Nonetheless, greater value lies in solving photocatalyst design and discovery challenges. A quick search on the ISI Web of Science found close to 71 000 publications related to photocatalysis stretching as far back as 1911. These are potentially a valuable source of data for ML training. Nevertheless, common issues that exist within the photocatalysis literature are the inconsistencies of data, originating from incomplete reported details, different reaction environment, and measurement protocols, which has made comparison of results between studies laborious. Further, the performance reporting practices (i.e., reaction rate, quantum efficiency, or solar-to-hydrogen efficiency) also differ from one research group to the other.[13] Therefore, utilization of standard test protocols and/or similar sources of illumination are crucial to ease the data extraction process.[14,15] The high dependency of light source (with various intensities and wavelengths) toward the photoactivity makes it one of the most important factors that should be standardized. In this regard, utilization of standard simulated solar irradiation (A.M. 1.5, defined by the American Society for Testing and Materials as the standard reference solar spectrum)[16] is considered the most practical approach. If light source with the same irradiation spectrum is employed, the difference in the amount of incident photons to the reactor can be easily normalized by measuring the internal quantum yields while taking into consideration the light irradiance calibrated using actinometry reagents (e.g., ferrioxalate).[17] For published photoactivity data measured under other light sources, the solar activity of the photocatalysts can still be inferred if the action spectra (for photocatalyst suspensions)[18] or incident-photon-to-photocurrent efficiencies (for photoelectrodes) are made available.[19] These measures would enable provision of consistent data to facilitate development of ML models, which at the current state seems intractable.

Given the current limitations as explained above, the foreseeable data set size with high quality and consistent information for the discovery of photocatalysts is at best in the range of thousands of data points. This demands a divergence from the common paradigm in ML, whereby a large data set is necessary to train an accurate ML model. One way to cope with a limited data set size is the integration of domain knowledge with data-driven ML model training. This photocatalysis domain-knowledge-driven ML framework could potentially be a more practical and astute approach for discovering efficient solar photocatalysts. In this Perspective, different categories of photocatalysis domain knowledge followed by recent data-driven ML developments from the heterogeneous catalysis viewpoint are summarized. Strategies governing the integration of domain knowledge into ML framework are proposed, and the potential efficacy of the framework for accelerated discovery of photocatalysts is discussed. As illustrated in Figure 1, coupling domain knowledge with data-driven ML at various stages supplements and complements the overall process of photocatalyst discovery.

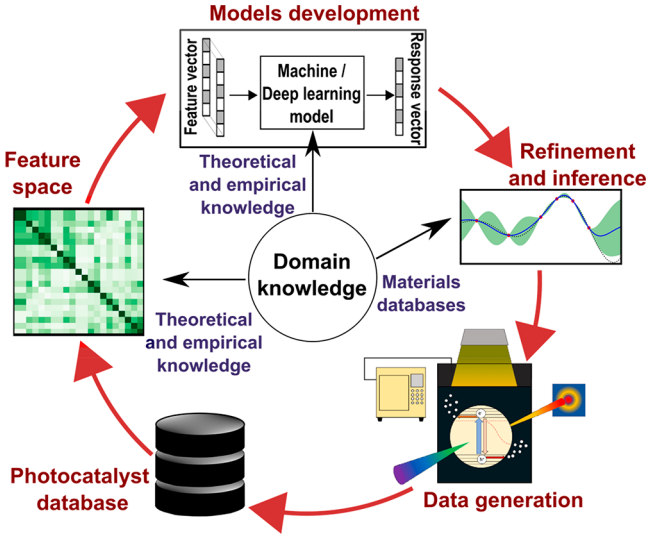


Figure 1. Integration of photocatalysis domain knowledge into ML framework that supplements and complements the overall process of photocatalyst discovery.

2. ESTABLISHING PHOTOCATALYSIS DOMAIN KNOWLEDGE

The term "domain knowledge" is used to refer to all current knowledge in the field (in this case, photocatalysis) based on existing theories, principles, and even experimental observations. Here, we classify photocatalysis domain knowledge into the following: (i) theoretical knowledge and empirical knowledge derived from first-principles and experimental observations, respectively; and (ii) knowledge from existing material databases.

2.1. Theoretical and Empirical Knowledge. As light-harvesting efficacy is one of the most important components governing the overall photocatalytic performance, the underlying physical model for solar photocatalysts shall begin with summing the amount of harvestable photons. By a conservative approach, where one active photon produces one electron–hole pair, the amount of absorbable solar photons ($N_{photon,SC}$) can be calculated from the standard solar spectrum (A.M. 1.5, 1 sun intensity) by integrating the fractional absorbed photons at an energy equal to or higher than the photocatalyst bandgap (E_g):

$$N_{photon,SC} = \int_{E_g}^{\infty} (1 - 10^{-t}) \frac{N_{photon,solar}}{E} dE \quad (1)$$

where α is the extinction coefficient of the photocatalyst at photon energy E, t is the compact thickness of the photocatalyst layer, and $N_{photon,solar}$ is the amount of solar photon at E. The first term within the integral is basically equivalent to the Beer–Lambert law for optical absorbance where $(1-10^{-A}) = (I_0 - I)/I_0$, or in other words, the fraction of photons absorbable by the photocatalyst at E, where $A = \alpha t$ is the absorbance, while I_0 and I are incident and transmitted (nonabsorbed) irradiation intensity, respectively. By adopting the Shockley–Queisser approach,[20] the unavoidable radiative recombination, RR(V), can be estimated:

$$RR(V) = e^{qV/k_BT} \cdot \frac{2\pi}{c^2h^3} \int_{E_g}^{\infty} \frac{E^2}{\exp[(E - qV)/k_BT] - 1} dE \quad (2)$$

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where q is the elementary charge, k_B is Boltzmann constant, T is the temperature, c is the speed of light, and h is the Planck constant. In the best case scenario under rapid surface charge transfer, equivalent to the short-circuit current ($V = 0$), the RR(0) can be easily estimated:

$$\mathrm{RR}(0) = \frac{2\pi}{c^2h^3}\int_{E_g}^{\infty}\frac{E^2}{\exp[E/k_BT] - 1}\mathrm{d}E \tag{3}$$

Hence, the maximum rate of extractable electron−hole pairs can be calculated from the photocharge balance:

$$N_{e^-\text{-}h^+} = N_{\text{photon,SC}} - \mathrm{RR}(0) \tag{4}$$

Note that the ideal bandgap for achieving the maximum rate of extractable electron−hole pairs is highly dependent on the irradiation spectrum (see eq 1-4), where in the current case of $N_{\text{photon,solar}}$ refers to the standard A.M. 1.5 (1 sun) solar irradiation.

While the above equations are exclusive for the bandgap excitation of semiconductors, there are cases where additional absorption arises from secondary light-sensitive components that does not lead to bandgap excitation (hence such components should be omitted from eq 1). These features, normally at lower energy than the absorption threshold of wide bandgap semiconductors, are not uncommon if the metallic components of d-band exists. This gives rise to the d−d transition that barely contributes to the overall solar harvesting, or in the special cases of plasmonic absorption, such as those from Au, Ag, Cu, and Pt, may lead to the generation of localized surface plasmon resonance (LSPR).[21−23] Typically, LSPR effects on modified photocatalysts (in multiple forms, ranging from deposited nanoparticles to core−shell structures) include the following: (1) hot (plasmon) electron injection, (2) local electric field that promotes photocharge separation, and (3) localized heat spots that aid organic degradation but may be detrimental to overall charge separation (since RR is a function of temperature, see eq 2).[24−26] These effects have been demonstrated to improve the overall photocatalytic and photoelectrocatalytic performances. Although the optical absorption model of small Pt nanoparticles has been proposed,[27] the observed phenomenon in plasmonic photocatalysis is yet to be comprehensible in any form of physical equations.

For the photocatalytic reaction to have a chance to take place, the photocharges must be able to diffuse to the surface in a way that the path length (L) taken by the photocharges must be less than the square root of the products of carrier diffusion coefficient (D_c) and lifetime (τ):

$$L = \sqrt{D_c\tau} \tag{5}$$

According to the Einstein relation, the diffusion coefficient and the charge mobility (μ) are related by the following equation:

$$\frac{D_c}{\mu} = \frac{k_BT}{q} \tag{6}$$

The carrier mobility is unique to each material as defined by its wave functions, as well as the associated carrier density and defects as defined by the dopant concentrations and the quality of the synthesized semiconductor crystal. The presence of electron scattering centers such as formation of impurities, defects, and phonons are inevitable in practical photocatalysts and result in the departure from its ideal state. Figure 2 shows the Boltzmann relation that correlates the carrier mobility with the carrier density of different materials.[28]

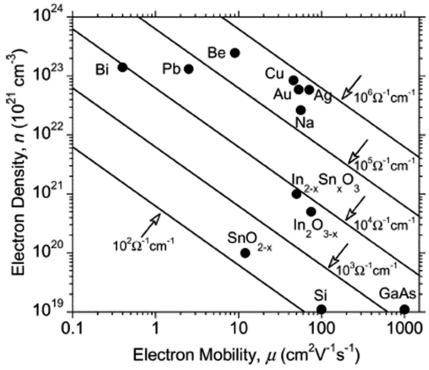


Figure 2. Boltzmann relation between electron density, electron mobility, and conductivity (represented as Ω^{-1} cm^{-1}) for a range of metals and semiconductor materials. Reproduced with permission from ref 28. Copyright 2004 The Royal Society of Chemistry.

While the Hall measurement is the standard physical technique for measuring carrier mobility, it can also be estimated from the first-principles calculations with input defects type and densities.[29] Spectroscopic techniques such as surface photovoltage spectroscopy (SPS),[30] intensity-modulated photocurrent/voltage spectroscopy (IMPS/IMPV),[31] and stepped-light induced measurements of photocurrent and photovoltage (SLIM-PCV)[32] have the advantage of simultaneously estimating both the D_c and τ. Like D_c, τ is also material-specific and subject to the influence of defects, whether in the form of charge trapping sites or recombination centers. In the literature, the reported values of L range from 2 to 4 nm for α-Fe_2O_3,[33] ∼100 nm for $BiVO_4$,[34,35] to 10 μm for TiO_2.[36] Ideally, the particle radius or film thickness shall be at a length scale ∼ L since any photocharges excited deeper below the surface would have little chance to reach the surface. This presumes that the light penetration depth, α^{-1}, is larger than L, which is generally true for most practical photocatalysts.

As the basic band potentials requirement, the photoelectron ought to have a more negative potential than that required for the reduction reaction, and vice versa for the photohole-mediated oxidation reaction (Table 1). The potentials of the photoelectron−hole pair can be taken at the conduction and valence band edge potentials, respectively:

$$E_g = E_{CB} - E_{VB} \tag{7}$$

Photoelectrons above conduction band edge (E_{CB}) and photoholes below valence band edge (E_{VB}) (i.e., hot carriers), produced by photonic excitations with energy higher than E_g, thermalizes rapidly to the respective edges. In the case of n-type photocatalyst

$$E_{CB} = E_f - \frac{\Delta E_f}{q} \tag{8}$$

where E_f is the Fermi level of the photocatalyst, ΔE_f is the energy gap between the Fermi level of the photocatalyst and E_{CB}. The ΔE_f can be estimated from

$$\Delta E_f = k_BT\ln\frac{N_D}{N_V} \tag{9}$$

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Table 1. List of Standard Reduction Potentials (E^0, V vs NHE at pH 0) of Some of the Most Common Photocatalytic Redox Reactions and the Associated Electrochemical Overpotentials Related to the Photocatalyst Surfaces or Attached Cocatalysts[37−46]

reduction reaction	standard reduction potential, E^0 (V vs NHE)	overpotential (V)	oxidation reaction	standard reduction potential, E^0 (V vs NHE)	overpotential (V)
Hydrogen Evolution Reaction (HER)			Oxygen Evolution Reaction (OER)		
$2H^+ + 2e^- \rightarrow H_2$	0.00	0.03 (Pt)	$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$(OER)	1.23	0.40 (Co_3O_4, NiO, RuO, RuO_2)
		0.12−0.25 (MoS_2)			0.50 (PtO_2, RhO_2)
		0.08 (CoP)			1.30 (TiO_2, MoO_2)
		0.15−0.21 (Mo_2C)			2.20 (VO_2)
		0.03−0.21 (N-carbon)			
CO_2 Reduction Reaction (CO_2RR)			Organic Oxidation Reaction (OOR)		
$CO_2 + e^- \rightarrow CO_2^-$	−1.49		$CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$	0.03	
$CO_2 + 2H^+ + 2e^- \rightarrow HCOOH$	−0.20	1.13 (carbon)	$CH_3OH \rightarrow \cdot CH_2OH + H^+ + e^-$	−0.95	
		0.20 (Pd/carbon)	$CH_3OH \rightarrow HCHO + 2H^+ + 2e^-$	0.23	
		0.52−1.22 (Sn)	$C_2H_5OH + 3H_2O \rightarrow 2CO_2 + 12H^+ + 12e^-$	0.08	
		0.33−0.88 (SnO_2)	$C_2H_5OH \rightarrow C_2H_4O + 2H^+ + 2e^-$	−0.20	
$CO_2 + 2H^+ + 2e^- \rightarrow CO + H_2O$	−0.11		$HCHO + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$	−0.08	
$CO_2 + 4H^+ + 4e^- \rightarrow HCHO + H_2O$	−0.08				
$CO_2 + 6H^+ + 6e^- \rightarrow CH_3OH + H_2O$	0.03				
$CO_2 + 8H^+ + 8e^- \rightarrow CH_4 + 2H_2O$	0.17				
$2CO_2 + 12H^+ + 12e^- \rightarrow C_2H_4 + 4H_2O$	−0.07				
N_2 Reduction Reaction (NRR)					
$N_2 + H^+ + e^- \rightarrow N_2H$	−3.20				
$N_2 + 4H^+ + 4e^- \rightarrow N_2H_4$	−0.36				
$N_2 + 6H^+ + 6e^- \rightarrow 2NH_3$	0.55				
Heavy Metal Reduction Reaction					
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	1.33				
Organic Reduction Reaction					
$RNO_2 + 6H^+ + 6e^- \rightarrow RNH_2 + H_2O$					

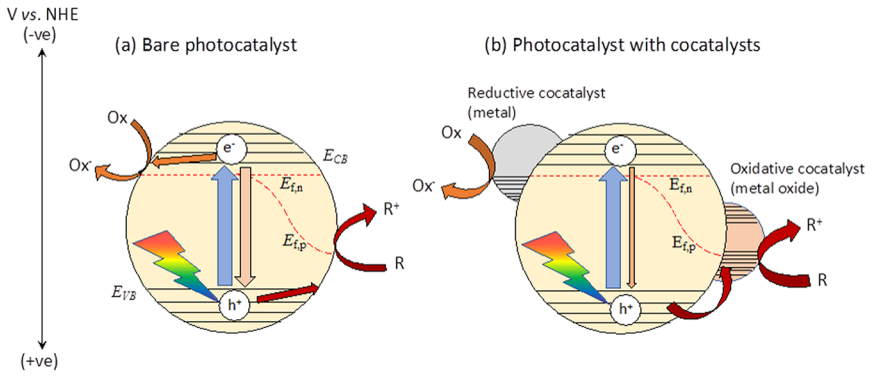


Figure 3. Comparisons of photocatalytic redox reactions taking place on (a) bare photocatalyst and (b) that loaded with reductive (metal) and oxidative (metal oxide) cocatalysts. The enhanced surface transfer on cocatalysts resulted in suppressed electron−hole recombination in (b).

where N_D is the donor density, while N_V can be further estimated from the effective mass of the electron (m_e*):

$$N_V = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{2/3} \quad (10)$$

The analogous eqs 8−10 can be written for *p*-type photocatalyst by replacing E_{CB} with E_{VB}, $-\Delta E_f$ with $+\Delta E_f$, N_D with acceptor density (N_A), and m_e* with the effective mass of hole (m_h*).

Besides fulfilling the band potentials requirement for the photocharges to have greater potentials than the target redox reactions, there are additional overpotentials associated with the reaction barriers. These overpotentials, which are at least a few millivolts, are specific to the surface of the photocatalysts and the target reactions.[47] It is a common practice to attach suitable cocatalysts on the photocatalysts surface if the latter is not sufficiently conducive to catalyze the target redox reactions. For example, the deposition of Pt cocatalyst (overpotential 0.03 V, Table 1) means that the E_{CB} needs only to be at −0.03 V vs NHE or more negative to catalyze the hydrogen evolution half reaction at an equivalent of XX mA $cm^{-2}{}_{cocatalyst}$. Likewise with deposited RuO_2 cocatalyst (overpotential 0.4, Table 1), the photocatalyst E_{VB} needs to be at least at +1.27 V vs NHE to catalyze the oxygen evolution half reaction at an equivalent of 1 mA $cm^{-2}{}_{cocatalyst}$. To ensure

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Table 2. Photocatalyst Bulk and Surface Properties Obtained from the Corresponding Measurement Techniques/Databases

sources		photocatalyst properties	techniques/databases
physical characterization	surface	surface adsorption energies	DFT calculation
		surface area	BET, ECSA
		surface defect density	XPS, Auger, EPR, Raman
		crystal facet	XRD, SEM, TEM, HRTEM
		surface potential	KPFM, AFM, STM, zeta potential
	bulk	charge lifetime, charge recombination rate	PL/TRPL, TRMC, TAS, THz/TRTS
		charge diffusion length	TRMC, PEIS
		charge mobility	TRMC
		charge density	MS, TRMC
		bandgap	UV−vis, PL, SPS
		band position/fermi level	MS, LSV, XPS valence, UPS
		composition/phase	XRD, ICP
		crystal structure/crystallinity	XRD
material databases	surface	surface adsorption energies	CMR, NREL, citrination, SCH
	bulk	band structure	CMR, NREL, AFLOW, MP, citrination
		crystal structure	CMR, MP, ICSD, citrination, COD
		effective electron/hole mass	CMR
		defect formation energies	NREL

proper band alignment and interfacial charge transfer, care must be taken that the E_{VB} of the photocatalyst lies less positively than that of the oxide cocatalyst to allow for interfacial hole transfer and that the Fermi level position of the metallic cocatalyst being more positive the photocatalyst E_{VB} to allow for interfacial electron transfer (Figure 3).

Apart from theoretical knowledge that is derived from first principles, domain knowledge can also come from additional observations on activity trends and mechanisms deduced from the experiments. This experimental knowledge that is inadequately described theoretically is regarded as empirical knowledge. This can be used to account for nonidealities, for instance, the formation of defects on photocatalyst prepared using different synthesis techniques that led to different activities even for the same semiconductor materials with identical macroproperties. In reality, there are inexhaustible types of defects at various length scales in the bulk as well as on the surface of each photocatalyst, ranging from oxygen or anionic vacancies, cationic vacancies, interstitial defects, line defects, defects at the grain boundaries, to twinning defects.[48,49] Establishing each type of defects and concentrations is highly challenging because many of these defects coexist within a photocatalyst particle, and their broad energy distributions overlap one another in an intricate fashion. Fortunately these defects usually vary systematically (within the same set of synthesis technique) as a function of physical properties, such as particle size and specific surface area, which are easily measurable thus allowing empirical relationship to be drawn.

To establish empirical domain knowledge, innumerable experiments are required to reliably prove the validity of the existing hypothesis. This form of prior knowledge that is yet to be associated with fundamental theories is of high significance, often because of the direct correlation with the desired outcome. Building upon the prevailing photocatalysis knowledge by continuous exploration within the area is crucial to enable accelerated photocatalyst discovery. For example, formation of defects on a photocatalyst is known to be detrimental because of the presence of charge trapping sites for electron−hole recombinations.[50,51] Further investigations, however, have shown that the presence of a surface defect, especially oxygen vacancies, that can be the active sites for specific reactions are therefore beneficial.[52,53] Crystal facet-engineering is another well-recognized approach that strongly influences the photoactivity, selectivity, and stability of the photocatalyst.[54−56] While the correlation between facet-dependent properties and photoactivity still could not be comprehended as established physical equations, a concrete understanding of these properties (i.e., surface electronic structure, atomic coordination and charge population)[57,58] has been manifested in numerous research experiments and stimulations.

Another important aspect in photocatalysis is the photostability of the catalyst, in which high stability is one of the prerequisite traits in achieving high performance. Previous studies have demonstrated that metal oxide (e.g., Cu_2O and ZnO) and metal sulfide (CdS and Cu_2S) experience extensive photocorrosion under illumination;[59−62] therefore, various measures are reported as potential solution.[63,64] In situ/operando characterizations probe dynamic changes of photocatalyst under irradiation and can help with insights on photostability (more details will be discussed in section 3.1).[65,66] Photocorrosion, another indicator of photostability, could influence bandgap and conductivity,[67,68] but universal theories to substantiate photocorrosion are not yet known. Although the fundamental theories regarding effects of defect formation, exposed crystal facets, and photocorrosion are still not fully understood, the prior studies provide interesting and useful knowledge to the photocatalysis research community, in terms of various bulk and surface properties that could potentially aid photocatalyst discovery.

2.2. Existing Materials Databases. Materials genome and crystallography databases are essential resources that provide previously measured or theoretically estimated materials properties necessary for executing the domain knowledge pipeline. Comprehensive databases bring great advantages as they could be used for screening of catalysts. Computational screening of materials for electrocatalytic the hydrogen evolution/oxygen reduction reaction,[69,70] thermal catalytic methanation,[71] and the photocatalytic water-splitting reaction[72] are some examples that demonstrated successful identification of new and efficient catalysts from the existing

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materials databases. In that respect, the Inorganic Crystal Structure Database (ICSD),[73] Computational Materials Repository (CMR),[74] AFLOW,[75] NREL Materials Database, Materials Project (MP),[76] Open Quantum Materials Database (OQMD),[77] Crystallography Open Database (COD),[78] National Institute of Science and Technology Materials Data Curation System (NIST),[79] Stanford Catalysis-Hub (SCH),[80] Materials Data Facility,[81] and PubChem[82] are some useful sources of physicochemical and quantum properties of large materials inventory. These extractable properties include wave functions, adsorption energies, band structures, crystal structures, and effective electron/hole masses (Table 2). Besides, the crystallography databases can also guide the selection of synthesizable materials (and potentially synthesis conditions) during the photocatalysts screening stage. Although only a fraction of these could serve as active photocatalysts, having knowledge of synthesizable materials could effectively narrow the search space of viable candidates.

3. MACHINE LEARNING

Machine learning broadly refers to techniques whereby a model of the relationship between some reference/input features (this term is generally used to refer to parameters that quantify the data of interest) and output parameters to be predicted from the input, is "learned" from a suitable training data set. In facial recognition systems, for instance, the relationship learned by ML models is between photographs of faces and the corresponding identity of people. During ML training, the model learns the significance of the features without being explicitly told (e.g., shape/position of eyes is more important than the background). After the training, the models can be used to infer the identity of a person given a new photograph that is not included in the training data. Similarly, in heterogeneous catalysis, the relationship modeled during training is typically between properties of catalyst as features and catalytic performance as output.[83−85] Once the models are trained, the activity of catalysts may be inferred without the need of conducting actual experiments/stimulations. In recent years, ML systems have started to be widely deployed in a range of applications, such as those in computer vision and speech processing, where the convergence of increasing computational power and availability of large training data sets has led to complex but powerful and accurate ML models. In general, the more complex the ML model being trained, the larger the training data set needs to be.

In the context of photocatalysis, the broad overview of data-driven ML accelerating discovery of active photocatalysts is illustrated in Figure 4. The gist of the process is that predictive ML models could offer an inexpensive route to determine the photocatalytic activity given the properties of the catalyst, thereby reducing the required efforts and resources when compared with typical approaches that are purely experimental and/or computational. The implementation of ML to accelerate photocatalyst screening requires a significant amount of training data to be acquired via reliable and standardized practices. Evaluation of trained models and subsequent refinement to improve accuracy with reduced experimental and computational overheads is also crucial. The associated components of a generic data-driven ML framework for photocatalyst design and discovery is explained herein.

3.1. Data Generation. Provision of a compatible data generation platform constitutes the first step in the implementation of ML framework for photocatalyst discovery. This component dictates the quality of data to be propagated to the ML models and thus influences the overall accuracy and reliability of the framework. Synthesis and characterization of various photocatalysts followed by their corresponding photocatalytic performance tests are parts of the experimental data generation procedure. Careful and consistent selection of the synthesis method is imperative to avoid additional discrepancy within the generated data. For instance, the same photocatalyst that is fabricated via different synthesis routes may seem to have similar basic chemical and optical properties, but distinct photoactivity due to the difference in properties is difficult to measure (i.e., defect types and defect densities). As a matter of fact, each synthesis method induces specific changes in the properties of the photocatalyst that is still not fully comprehensible. Therefore, standardized synthesis and photoactivity test protocols within a controlled environment are required during data generation, unless the structure−property−activity relationship could be proven to be independent of the synthesis procedures. Otherwise, representation of materials as model inputs becomes a challenge. Selection of structural and property attributes in feature space should be made to account for variability in synthesis protocols. If not, a flexible model structure, or embedding known correlations in feature space, might enable to "learn" such information from data (will be discussed in sections 4.1 and 4.2).

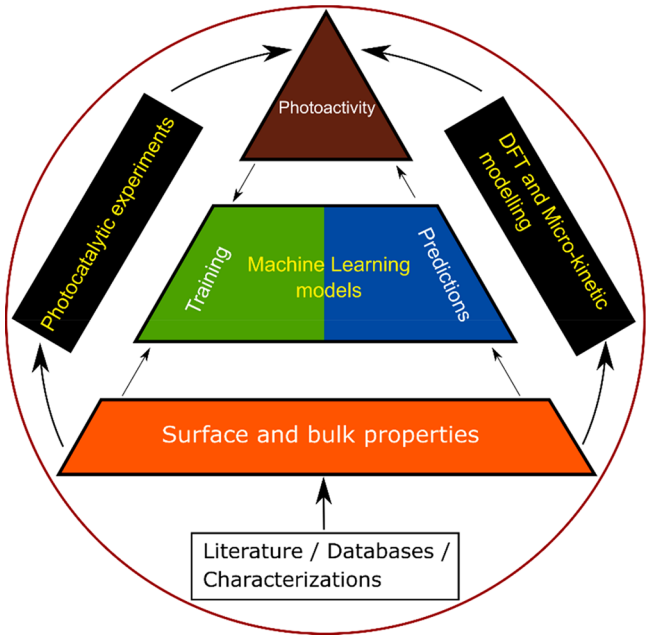


Figure 4. Execution of ML as an accelerated property-activity determination route. With successful training of accurate predictive models, reliance on experiments and stimulations is significantly reduced, thereby enabling robust screening of photocatalysts.

As elaborated in section 1, presuming that photocatalytic data in the open literature is insufficient or unsuitable to be extracted for ML, it becomes imperative to generate new data sets in a consistent manner, preferably from the same source. Although a wide range of photocatalyst synthesis methods, such as hydrothermal, wet impregnation, sol−gel method, chemical precipitation, and flame spray pyrolysis are available, these methods are primarily low-throughput approaches that require laborious efforts and resources. Given this, high-

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throughput and combinatorial techniques are valued choices for generating decent-sized data sets.[86,87] Among the established techniques, electrodeposition,[88,89] combinatorial inkjet printing,[90,91] and modified sol−gel method[92,93] are some of the most common methods used to produce wide arrays of photocatalytic materials for high-throughput screening. This should be complemented by spatially resolved characterization techniques, as illustrated in Figure 5, to obtain various

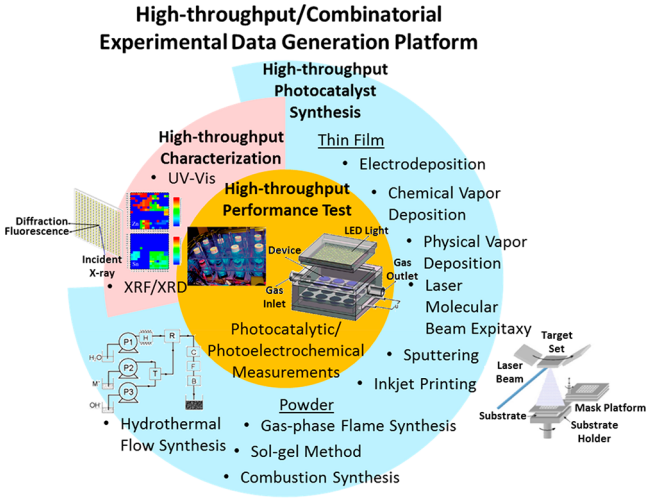


Figure 5. High-throughput/combinatorial photocatalyst syntheses, characterizations, and photocatalytic/photoelectrochemical performance tests that are available in the literature. Reprinted with permissions from ref 96. Copyright 2013 Elsevier. Reproduced with permission from ref 97. Copyright 2005 American Chemical Society. Reproduced with permission from ref 98. Copyright 2007 American Chemical Society. Reproduced with permission from ref 99. Copyright 2010 American Chemical Society.

physicochemical information on the photocatalyst arrays, including a wide range of photocatalytic and photoelectrochemical measurements, glancing angle or wide-angle X-ray/neutron scattering, Raman microscopy, and X-ray absorption spectroscopy. Similarly, high-throughput first-principles computational methods have been used to screen nitrides/oxynitride and two-dimensional photocatalysts for the photocatalytic water splitting reaction based on optical and electronic properties.[94,95] In consideration of utilizing photocatalysis data in ML, high-throughput computational studies could also be employed to generate a large volume of reliable training data.

An important aspect not to be neglected during data generation stage is to monitor the dynamic changes of photocatalyst during the reaction. In principle, the most desirable photocatalyst is a stable material that does not undergo any form of changes during the reactions. Nonetheless, the changes in oxidation states, morphologies, and/or compositions of photocatalysts or cocatalysts under reactive environment (upon illumination) are hard to be prevented, especially for materials that are susceptible to photocorrosion. As mentioned in the previous section, in situ/operando characterization techniques could also be employed to understand and mitigate the photocorrosion of semiconductors. In addition, to monitor the photocorrosion pathways, these advanced techniques also enable the understanding of reaction mechanisms.[100,101] A recent study has utilized in situ synchrotron X-ray absorption spectroscopy to probe the changes of copper species during CO_2 photoreduction.[102] The transformation of Cu(II) species to Cu(I)/Cu(0) during the reactions was shown to enhance the photocatalytic reduction of CO_2 to CH_4. A similar phenomenon was also observed in another study when PtO_2 was reduced to active Pt metal during gas phase photocatalytic hydrogen generation reaction.[103] These studies demonstrated that some photocatalysts are primed to be more active under illumination during the reactions. Thus, incorporation of data generated from in situ/operando characterizations could be helpful for the ML model to learn the actual state of the photocatalyst during the reactions.

The generation of large volumes of reliable and consistent data from authentic sources could be useful for not only the ML framework but also open-access database formulation. Sharing knowledge with the global photocatalysis community provides a broad impact on shaping future of photocatalysis research. For one, a photocatalysis-specific database could enable sharing of invaluable data/information in a consistent and standardized manner, mitigating bottlenecks explained in section 1. Second, data amalgamated into common machine-readable formats could be conveniently mined and queried, shared, and explored to construct models or finding patterns in the data.[104] This offers long-term benefits to the community in terms of harnessing available data with tailored ML algorithms to enable new discoveries rather than starting from scratch and consuming unnecessary time and resources. The database could greatly shift the focus toward resourceful data analytics rather than concern about the availability of consistent data. Because photocatalysts share some common traits (e.g., band gap, Fermi energy, d-band center, etc.) from other areas underpinning heterogeneous catalysis and material sciences, such an "e-collaboration" platform of rapid data transfer and communication would be valuable largely for materials and catalysts informatics.

3.2. Feature Engineering. Each training data point (i.e., photocatalyst properties and its photocatalytic activity) potentially includes extensive information on the material's physicochemical and quantum characteristics that are probed or calculated experimentally or extracted from materials databases (Table 2). In other words, representation of a certain material in feature space entails inputs from physical characterizations and/or material databases, which subsequently governs the level of detail/accuracy assimilated in the ML model. This raw data could potentially be noisy or complex with hidden relationships between each input property, which in turn could make the direct inference of photocatalyst activity challenging. Thus, data needs to be preprocessed before it is fed to the ML engine, which often involves constructing machine readable features, evaluating sensitivity and selecting relevant features on the basis of the relative significance of each feature. Identification of a relevant and suitable number of features is crucial for achieving a good bias-variance trade-off when training ML models. The higher the dimensionality of the features, the larger the training data set needs to be in order to obtain reliable estimates of model parameters, thus avoiding overfitting of the model.[105] On the other hand, using an insufficient number of features leads to underfitting, which in turn also leads to poor predictive performance. This essentially requires assessment and dropping of features, a concept widely implemented in photodegradation reactions to evaluate importance of parameters such as light intensity, catalyst loading, solution pH,

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irradiation time, and reactant concentration, toward the photoactivity.[106,107] Hence, considering inexhaustive photocatalysis feature space (some key photocatalyst properties are included in Table 2), feature evaluation and reduction is imperative for efficient development of ML models.

A second (but somewhat related) goal of feature engineering is to deal with hidden correlations present in the complex raw information (i.e., the various photocatalyst properties). These correlations may exist between various properties of photocatalysts, which could potentially be descriptors of the photocatalytic activity, some of which may be obvious based on human intuition, while others are not. Taking the example of surface potential that affects the charge transfer and surface catalytic properties, the current approach to obtain the value is by ab initio techniques, such as solving the Kohn−Sham equations in DFT. If ML could identify a complex interlink between various materials-related features (e.g., conduction/valence band edge, Fermi level, density of states, effective mass of electrons, donor density, and permittivity), the formulated correlation or descriptor could be representative of the photocatalytic activity in a similar manner that the surface potential would. This way, computationally demanding calculations could be bypassed in ML. Similarly, adsorption energies of reactant/intermediate/transition states for a large range of materials might be estimated by ML and used as the activity descriptors. However, formulating correlations this way would not convey any physical or chemical meaning of the connections between features. The open question is whether domain knowledge could be leveraged to identify such compact representation between features as physically meaningful descriptor of photocatalytic activity.

Raw photocatalysis data that can be collected in accordance with Table 2 might contain a mix of quantitative and qualitative information. The manner in which information is represented could affect the efficacy of learning process of ML models.[108] The problem offers grounds for tapping domain-knowledge to design fingerprints for knowledge-based representation of raw scientific information into machine-readable form. Fingerprinting information, such as chemical or molecular structure of species, active sites for adsorption, and environment surrounding adsorption sites of species, is particularly challenging, but there are solutions available. A Coulomb matrix is one of the examples fingerprinted on the basis of the Cartesian coordinates and nuclear charges of atoms, which are frequently used as features for describing various thermodynamic properties of molecules and solids.[109] Meanwhile, a simplified molecular-input line-entry system (SMILES) representation can fragmentize the chemical structure of adsorbed species to be stacked into a feature vector, as benchmarked by accurate prediction of formation energies for surface adsorbates in syngas reaction on Rh(111).[110] A partial radial distribution function[111] for describing electronic properties of crystal based on pairwise interactions between atoms and property-labeled materials fragments[112] based on molecular network are some other examples. These examples are indicative that feature engineering could help encoding complex scientific data, irrespective of the scale of probed information, as physically relevant features for ML applications.

3.3. Development and Refinement of Models. The training and refinement of ML models is essentially the estimation of a function that maps the chosen inputs (features) to the required outputs (e.g., photocatalytic activity). Generally, this is carried out by choosing a suitable family of parametric functions defined over the feature space (obtained from feature engineering) and employing suitable training algorithms to infer appropriate model parameters from the training data (obtained from the data generation platform). Once the model is inferred, photocatalytic activity corresponding to previously unseen regions of the feature space can be estimated and used as the basis for identifying new photocatalysts. The efficacy of this approach would depend on the ability of the ML model to generalize from the training data to unseen regions of the feature space. The initial choice of the family of functions chosen as the ML model leads to the different model architectures, the commonly employed ones are artificial neural networks (ANN),[113] decision trees,[114] support vector machines,[115] and naive Bayes classifiers.[116] The training mechanism of model coefficients also differs because of variations in the algorithmic description. For instance, ANNs comprise several interconnected neurons arranged in input, hidden, and output layers. Each connection carries a certain weight corresponding to the strength of the connection, which is optimized during training by the backpropagation process. Meanwhile, decision trees exhibit a transparent architecture with nodes as attributes which are split on the basis of heuristics leading to a final outcome—either a class label or a value depending on classification or regression problem, respectively. Support vector machines, on the other hand, apply transformation to the data by projecting from low-dimensional to high-dimensional space using a kernel function. A summary of supervised learning tools for materials design and discovery is provided by Liu et al.[117]

Several examples associated with the aforementioned supervised learning can be found in the photocatalysis literature. Eskandarloo et al.[118] trained an ANN to model the photocatalytic activity of TiO_2/NiO nanoparticles for degradation of triphenylmethane dye in the presence of various additives. Similarly, Fathinia et al.[119] employed supervised learning via ANN to predict kinetics of photocatalytic ozonation of phenazopyridine on TiO_2. Another example is photoreforming of glycerol where process parameters, such as glycerol concentration, TiO_2 loading, amount of cocatalyst, and pH of solution, were optimized with help of ANN.[120] Irrespective of the variation in applications, mechanics of training of these models was broadly the same. Specifically, input features were mapped with targeted outputs using an ANN architecture, where coefficients/synapses were trained with the back-propagation technique. The technique is highly specific to ANN albeit the values for hyperparameters were different in these cases due to variation in nature as well as quality of data sets used to train the models. Given the choice of a broad spectrum of ML models mentioned above, suitability of ANN for scarce data sets is actively questioned because of overfitting concerns. This dilemma is prevalent when opting for a complex model, with multiple hidden layers containing several neurons, for high-dimensional data sets with a sample size of a few hundred.

In applications where large labeled data sets (i.e., inputs with known outputs) are available to be taken to represent the entire feature space, the typical approach would be to hold out a portion as a test set, train the ML models with the remaining data (training set), and then validate the models on the held out test set to provide an indication of real-world performance. However, the costs involved with obtaining labeled data and the consequent data scarcity in photocatalysis (and more

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broadly in other similar physical sciences) calls for a different strategy. Specifically, approaches in the vein of active learning and continuous model refinement could be more practical. These techniques presume that data representative of a fraction of the feature space is readily available which enables strategic and iterative exploration of the rest of the feature space. A representation of active ML in context of photocatalysts discovery is provided in Figure 6.

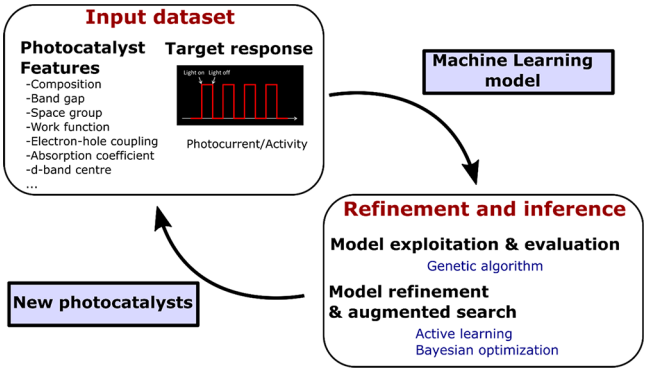


Figure 6. Model refinement and optimization loop for accelerated discovery of novel photocatalysts by iteratively establishing new query points with ML for potential experiments.

Active learning refers to a class of ML algorithms that can be employed where unlabeled data (i.e., input features with unexplored outputs, such as activity) can be availed relatively easily, but labeling requires significant effort. Strategies classified under active learning could allow continuous exploration of the space of materials by suggesting new photocatalysts to be evaluated. The obtained new input data could then enable further refinement of the predictive ML model. Some examples of this technique could be seen in terms of identification of bimetallic surfaces with optimum energetics for CO_2 reduction and H_2 evolution reactions[121] and discovery of $BaTiO_3$-based piezoelectrics with high electrostrain.[10] Bayesian Optimization[122] is another approach which excels in materials discovery problems as a guided-search tool in comparison with conducting random or exhaustive search. The approach is benchmarked by successful prediction of new ternary oxides,[123] as well as thermodynamically stable ternary compounds requiring fewer resources compared to first-principles calculations.[124] Genetic algorithms[120,125] are another class of optimization algorithms that might help identifying optimal photocatalyst candidates in discrete feature space. Inaccuracies in trained ML models might impede viability of the solution inferred from genetic algorithms; thus, possibilities for model refinement should be kept open. Contemplating out-of-the-box approaches from the class of feedback control-based learning optimizer one may be drawn toward reinforcement learning,[126] made appealing by recent accomplishments in solving complex problems, such as the game of Go.[127] Though the idea is catching attention in the domain of chemical reaction optimization using computer simulations,[128] it may be premature to advocate the use in data-limited field of photocatalysis at the current stage.

3.4. Challenges with Data-Driven Approach. While the approach outlined in the previous section appears rational in principle, several challenges arise with implementation. For one, data generation platforms may not provide enough samples to sufficiently populate the selected feature space, which in turn could significantly hamper the accuracy of the ML models leading to flawed suggestions during the iterative model refinement stage. The ability of the data-driven ML models in this framework to generalize could also be in question because of the absence of validation data, particularly when extrapolating the models beyond the bounds of the training data set. Lack of interpretability of the models could also lead to reliability concerns. Finally, commonly employed ML models tend to be continuous functions on the feature space which could lead to suggested "new photocatalysts" having chemical compositions which are potentially irrational or impossible to synthesize. To cope with the mentioned challenges, integration of photocatalysis domain knowledge (refer to section 2) with the presented data-driven ML approach would be desirable for accelerated solar photocatalyst discovery.

4. INTEGRATING DOMAIN KNOWLEDGE IN MACHINE LEARNING

Human domain knowledge comprises an understanding of fundamental causal relationships and laws obtained after years of experience from theoretical and experimental viewpoint and is very distinct from ML in the manner in which it leads to predictions. In particular, ML is highly effective at pattern identification and matching from large, high dimensional data sets in the absence of underlying knowledge (i.e., human-like deeper understanding of causality). For instance, ML can detect an increase in cocatalyst loading to positively correlate with alcohol reforming performance without tangibly reasoning the superior activity.[120] Since ML provides necessary data-centric tools to approach the problem of photocatalyst discovery, adaptation of ML capabilities to resonate with knowledge of photocatalysis is of interest. The concept of "knowledge discovery"[129,130] was propagated earlier for heterogeneous catalysis, which set the precedence, but has not yet been adopted by the photocatalysis community. Consequently, we suggest integration of photocatalysis mechanistic understanding into data science models to close the cognition-gap and to achieve an optimization loop with potential to accelerate discovery of novel photocatalysts.

In general, there are a myriad of ways by which domain knowledge informs the design and the development of ML systems, ranging from strategies for training data collection, to the choice of modeling methodologies, training algorithms, and inference methods. Within the photocatalysis research space, we expect to be able to make use of domain knowledge, both directly and indirectly, to allow for complex models to be developed with the available data while reducing the risk of overfitting (which would be less of a risk in fields employing ML where large training data sets may be available). Specifically, we propose three approaches (as illustrated in Figure 1) in this section that may allow for this.

4.1. Structural: Within ML Models. Typically, ML models are chosen from a broad class of flexible functions that can fit any arbitrary mapping via an appropriate choice of function parameters (which are estimated during ML model training). However, we are not constrained to using such functions and may instead choose a model that is a combination of a deterministic component based on domain knowledge (such as models described in section 2.1, based on fundamentals of physics from first-principles) and a stochastic component that extends the model toward realistic nonideal conditions by learning new relationships present in the training

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data but not available in the theoretical models obtained from first principles. This way, ML would develop the capability to compare the deviation of theoretical model that defines the structure−property−activity relationship with experimental data, making it possible to capture veiled structures that are either difficult to measure or in some cases impossible to measure. The structural integration of domain knowledge supports the inclusion of model validation protocols (e.g., k-fold cross validation, bootstrapping, hold-out method, etc.) to supplement generalization capabilities by mitigating overfitting risks.

4.2. Geometric: Embedded in the Feature Space. Almost all ML algorithms rely on knowledge of how similarity between data points can be measured, either by defining a suitable inner product on the feature space or more broadly a metric on the feature space. Under certain circumstances these metrics can also be learned from appropriate training data sets.[131] In the context of photocatalysis, this provides a suitable venue for incorporating potentially rich domain knowledge to augment data-driven learning. For instance, if the correlations between two or more properties/features in our feature space is known to the photocatalysis community (e.g., if crystallite size/crystallinity of the photocatalyst is in direct relationship with its bulk defect density), this information could be incorporated into the distance metric employed by the ML model. We expect that graph-based models could be employed to learn covariance structures across the feature space (in the case of undirected graphs) as well as more complex relationships (with directed graphs).[132]

4.3. Strategic: Utilizing Materials Databases. A third approach to incorporating domain knowledge makes use of community databases in the form of known materials. Specifically, gearing the ML models toward the prediction of known materials rather than searching blindly in exhaustive feature space, that is, transposing the machine inference to be carried out in discrete domain rather than a continuous one, where the former would ensure that photocatalysts suggested by ML could indeed be practically synthesized in lab. In terms of implementation, by imposing suitable constraints on the feature space and the metrics defined on it, we could potentially reduce the hypothesis space that needs to be searched and consequently enhance the photocatalyst discovery process. A range of untested photocatalyst candidates from the large inorganic materials libraries such as the ICSD, COD, citrination, MP, and NREL Materials Database could be incorporated as domain knowledge in the heuristics space for optimizations.

5. SUMMARY

With the recent emergence of ML systems to facilitate the design and discovery of heterogeneous catalysts, significant opportunities also exist in accelerating photocatalyst discovery. Factors responsible for lack of adoption of ML in photocatalysis are the state of available data (insufficient volume of reliable data, incomplete detail in reported literature, inconsistencies in measurement protocols, nonstandardized experimentation equipment) and the challenges with acquiring new data (slow and expensive experiments/simulations, lack of direction). The predicament of scarce data sets impedes the execution of machine learning algorithms which are inherently data-driven, and thus, reliability and interpretability of models are frequently challenged. We foresee the coupling of photocatalysis domain knowledge with machine learning as a viable attempt to mitigate the concerns. We have conceptualized a multicomponent workflow leveraging capabilities of data-driven ML to accelerate discovery of solar photocatalysts on combination with domain knowledge. The key proposition involves training ML models, supplemented by human expertise, exhibiting ample complexity to capture intricacies underpinning the mechanistic of photocatalysis. Constraining model predictions with existing knowledge could subsequently lead toward active solar photocatalysts with rationally synthesizable compositions. The fusion of ML and photocatalysis knowledge might pave the way toward development of robust catalyst screening platform greatly benefiting photocatalysis community.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.9b02531.

Appendix: abbreviations used (PDF)

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Notes

The authors declare no competing financial interest.

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example, the >-Fe_2O_3 or WO_3 with E_{CB} being less negative than the hydrogen evolution potential, or NiO with E_{VB} being less positive than the oxygen evolution potential, can be used for the PEC water splitting applications under applied bias.

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